

Statistical Volatility of Real Metal Commodity Prices: A Long-Run Sub-sampling Analysis, 1901-2020

Ketu Gugushvili (I6357186)

Date: October 15, 2026

1. Introduction

Metals are a group of materials with distinctive chemical properties that show prominence in numerous fields such as scientific research, construction, industrial production, and electricity [1]. Commodity metals, which are obtained through mineral extraction, are often cited for their economic influence and substantial price fluctuations over time [2, 1]. To assess price fluctuations, it is important to consider both price levels and volatility. Price levels describe commodity costs relative to their specific time frame, and price volatility estimates variability across these price changes. Academic studies on this topic examine long-run real, inflation-indexed commodity prices and corresponding fluctuation trends [3].

The research question of this report is: Which of the most prevalent metals experienced the greatest real-price volatility between 1901 and 1010, and how has their volatility changed over time? The analysis will investigate 11 metals from the dataset taken from Our World in Data 4. The aim of the report is to achieve this by answering 3 main objectives: 1. Rank metals by their overall estimated volatility, 2. Compare volatility between 1901-1959 and 1960-2020, 3. Investigate volatility changes using a non-overlapping 10-year subsample. The precision of the results of each analysis will be evaluated by calculating statistical uncertainty.

2. Data sample

The dataset was obtained from Our World in Data [4], with historical commodity-price data attributed to Jacks (2019) [3]. It contains annual inflation-adjusted price indices for 11 metals: iron ore, bauxite, tin, zinc, steel, manganese, aluminum, chromium, copper, lead, and nickel. The common period is 1900–2020, with 1900 = 100. Each metal has 121 annual index observations, producing 120 annual percentage changes.

3. Methodology

3.1 Annual percentage changes

Annual percentage changes were calculated from the real price indices using:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100 \tag{1}$$

where P_t is the real price index in year t and r_t is the annual percentage change. The first annual change corresponds to 1901.

3.2 Statistical volatility

Volatility was measured using the sample standard deviation:

$$s = \sqrt{\frac{1}{N-1}\sum_{t=1}^{N}(r_t - \bar{r})^2} \tag{2}$$

where N is the number of annual changes and $\bar{r}$ is their mean. Higher values of s indicate greater variability in annual percentage changes.

3.3 Statistical uncertainty

The approximate standard error of the estimated standard deviation was calculated as:

$$SE(s) \approx \frac{s}{\sqrt{2(N-1)}} \tag{3}$$

Error bars represent ± 1 approximate standard error. This approximation assumes independent, approximately normally distributed observations [5].

3.4 Historical sub-sampling and distribution shape

Volatility was compared between 1901–1959 ($N = 59$) and 1960–2020 ($N = 61$). The relative percentage change was calculated using:

$$\Delta s(\%) = \frac{s_2 - s_1}{s_1} \times 100 \tag{4}$$

The 120 annual changes were also divided into 12 non-overlapping 10-year subsamples. Volatility and its approximate uncertainty were calculated for each metal and subsample, while skewness and excess kurtosis were used for investigating asymmetry of the distribution and tail characteristics [6].

4. Results and discussion

4.1 Overall volatility across metals

Zinc had the highest estimated volatility (25.50 ± 1.65 percentage points), which was followed by manganese (25.41 ± 1.65). Bauxite had the lowest estimate (13.64 ± 0.88) and, notably, Zinc's estimated volatility was approximately 1.87 times larger than bauxite. The difference found between zinc and manganese was small relative to their approximate uncertainties.

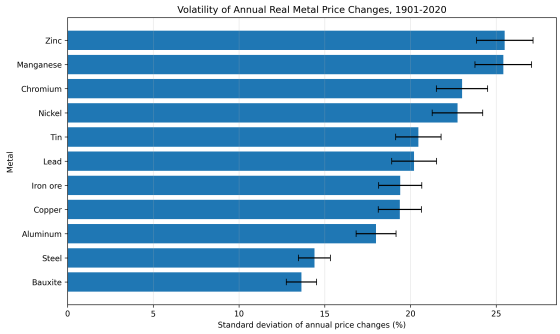


Figure 1: Estimated volatility of annual real metal-price changes, 1901–2020. Error bars represent ±1 approximate standard error.

4.2 Changes in volatility over time

The estimated volatility of Nickel displayed an increase from 8.27% to 30.82% (+273%), while iron ore decreased from 24.57% to 12.74% (-48%). Chromium increased from 19.37% to 26.14% (+35%). These results indicate that volatility changes and trends varied across different metals.

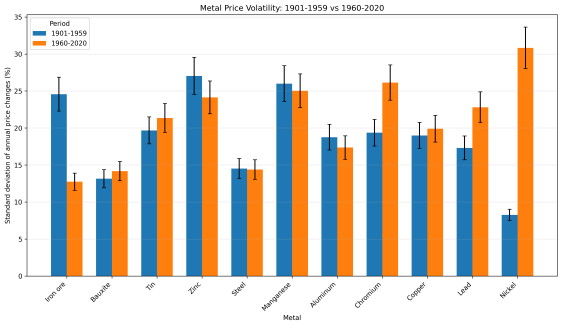


Figure 2: Estimated volatility in 1901–1959 and 1960–2020, with ±1 approximate standard error.

4.3 Sub-sampling and distribution analysis

The decadal analysis calculated high volatility for nickel in 1981–1990 (59.7%), zinc in 1911–1920 (56.3%), and iron ore in 1931–1940 (55.9%). These values are larger relative to their full-period estimates, indicating a significant difference in volatility across historical subsamples. However, each decade contained only 10 observations.

Zinc exhibited positive skewness (2.95) and high excess kurtosis (15.08), indicating an asymmetric, heavy-tailed distribution [6]. Therefore, it is not guaranteed that the normality assumption underlying the approximate standard errors will hold.

5. Conclusion

Metal-price volatility differed across commodities and historical periods. Zinc and manganese had the highest full-period estimates, while bauxite had the lowest. Nickel showed the largest proportional increase between the two historical periods, while iron ore showed the largest decrease. Decadal subsampling revealed additional time frames of high volatility. However, small subsamples and distributional assumptions limit the precision of the uncertainty estimates.

References

[1] Rabah Arezki and Akito Matsumoto. Metal prices signal global economic shifts. In Rabah Arezki and Akito Matsumoto, editors, *Shifting Commodity Markets in a Globalized World*, chapter 5. International Monetary Fund, Washington, DC, 2017. URL `https://www.elibrary.imf.org/display/book/9781484310328/ch005.xml`.

[2] W. J. Rankin. Sustainability. In S. Seetharaman, editor, *Treatise on Process Metallurgy*, pages 1376–1424. Elsevier, Boston, 2014. doi: 10.1016/B978-0-08-096988-6.00033-X.

[3] David S. Jacks. From boom to bust: A typology of real commodity prices in the long run. *Cliometrica*, 13: 201–220, 2019. doi: 10.1007/s11698-018-0173-5.

[4] Our World in Data. Real commodity price index, metals. `https://ourworldindata.org/grapher/real-commodity-price-index-metals`. Data originally sourced from Jacks (2019). Accessed 15 October 2026.

[5] NIST/SEMATECH. Engineering statistics handbook: Confidence limits for standard deviation. `https://www.itl.nist.gov/div898/handbook/prc/section2/prc231.htm`, . National Institute of Standards and Technology.

[6] NIST/SEMATECH. Engineering statistics handbook: Measures of skewness and kurtosis. `https://www.itl.nist.gov/div898/handbook/eda/section3/eda35b.htm`, . National Institute of Standards and Technology.